D. Details of GSR-Bench

1) Simulation Infrastructure: GSR-Bench is built upon a lightweight, web-native simulation framework designed specifically for evaluating high-level embodied reasoning, spatial understanding, and decision-making. Unlike traditional simulators that focus on low-level motor control, GSR-Bench emphasises the validation of an agent's ability to interpret complex scenes and plan long-horizon tasks.

To support this, the benchmark provides a rich set of 63 distinct object types, ranging from simple geometric primitives to complex articulated furniture. These objects are interconnected through a set of spatial and logical relations (e.g., OnTop, Inside, IsOpened) that form the basis of the scene graph. A summary of the detailed asset library and defined state space can be found in Table VIII.

TABLE VIII: GSR-Bench Asset Library

Category	Count	Typical Examples
Basic Primitives	11	Boxes (Red, Yellow, Blue), Mugs (Red, Yellow, Blue), Cubes (Red, Yellow, Blue, Green, White)
Articulated Assets	10	Movable furniture and appliances, including Cabinets with multiple drawers and Refrigerators, and Boxes with functional lids.
Stable HOPE	14	Standardized food and condiment items: Milk cartons, Ketchup, and Mustard bottles.
Stable Scanned	11	High-fidelity scanned kitchenware: Plates, Bowls (White/Red), and Trays.
Turbosquid	17	Diverse commercial 3D models: Coffee machines and specialized kitchen appliances.
Total Assets	**63**	A diverse collection for evaluating high-level task planning and decision-making.

And the technical architecture of GSR-Bench is designed to bridge high-level semantic planning with physically environment states through the following three components:

- **Physics and Rendering**: The simulator employs cannones for rigid-body dynamics (gravity, friction, and collisions) to ensure physical plausibility. Three.js provides WebGL rendering, offering a high-throughput, browser-based alternative to heavy simulators like Isaac Gym for large-scale data synthesis.
- **Communication and Action Interface**: Communication and Semantic Interface: ROS2 serves as the communication middleware via rosbridge suite. ROSManager orchestrates the system by broadcasting a 10Hz State Summary and listening for high-level semantic commands (e.g., move [A] to [B]). To focus on reasoning, the simulator directly updates object states upon receiving commands, effectively decoupling task planning from robot kinematics.
- **Real-time Scene Graph Extraction**: Scene graphs are dynamically generated from the ground-truth physics state. Spatial relations (e.g., OnTop, Inside) are computed via AABB (Axis-Aligned Bounding Box), while logical states (e.g., is_opened) are extracted directly from articulated object attributes.

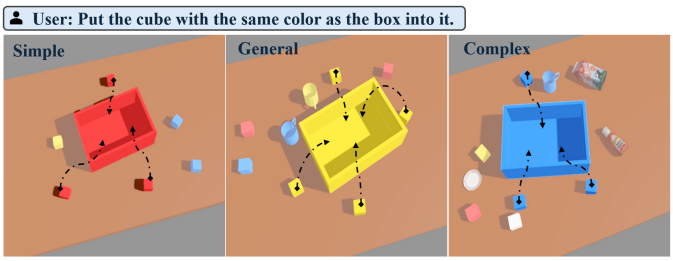

Fig. 13: Examples of semantic object disambiguation tasks

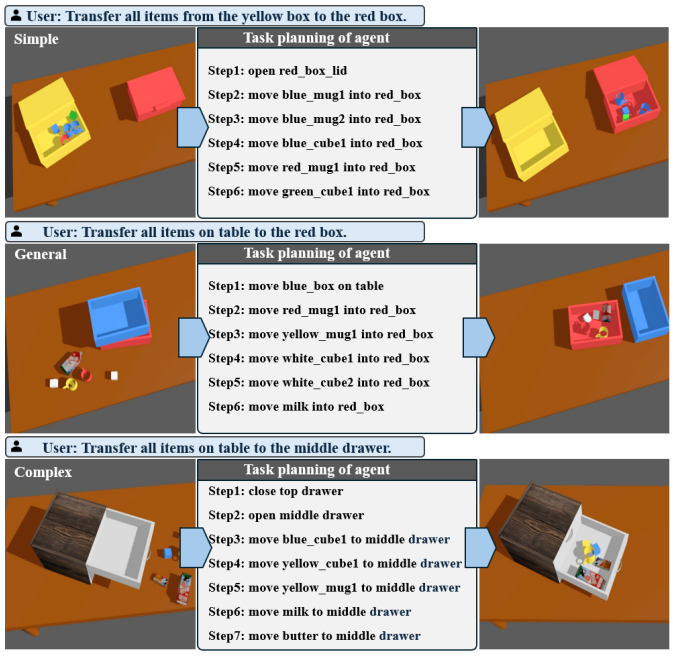

Fig. 14: Examples of spatial-aware sequencing tasks

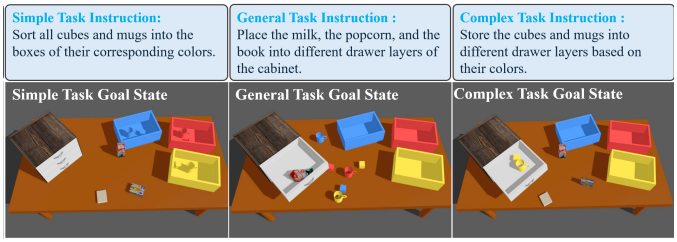

Fig. 15: Examples of goal-conditioned generalization tasks

2) Task Suites: GSR-Bench presents a comprehensive evaluation scheme intended to assess an embodied agent's performance across three fundamental cognitive domains: semantic object disambiguation, spatial-aware sequencing, and goal-conditioned generalization. Each suite comprises three levels of increasing difficulty, which are achieved by scaling up the number of obstacles, increasing physical constraints or broadening the scope of the instructions. The specific natural language instructions for each test are summarized in Table

IX. The detailed environmental configurations for each suite are described below:

a) Semantic Object Disambiguation: This suite assesses an agent's ability to identify target nodes within scene maps containing diverse object categories. The complexity of the scenes increases with higher densities of interfering elements and wider object category diversity.

- **Easy**: The scene contains one box (randomly Red, Yellow, or Blue) and 7–10 cubes of various colors on the table. *Instruction: "Pick up all cubes that have the same color as the box and put them inside the box."*
- **General**: The distractor density increases with 9–12 cubes and mugs of various colors added to the scene. *Instruction: "Pick up all cubes that have the same color as the box and put them inside the box."*
- **Complex**: In addition to the items at General level, there are three randomly selected items of a different type (e.g., plate, bowl, milk) are introduced. *Instruction: "Pick up all cubes that have the same color as the box and put them inside the box."*

TABLE IX: Instruction Specification across GSR-Bench Task Suites

Task Suite	Level	Natural Language Instruction
Semantic Object Disambiguation	Simple General Complex	"Pick up all cubes that have the same color as the box and put them inside the box."
Spatial-Aware Sequencing	Simple	"Transfer all items from the yellow box to the red box."
	General	"Move all objects on the table into the red box."
	Complex	"Transfer all objects from the box into the middle drawer."
Goal-conditioned Generalization	Simple	"Sort all cubes and mugs into the boxes of their corresponding colors."
	General	"Place the milk, the popcorn, and the book into different drawer layers of the cabinet."
	Complex	"Store the cubes and mugs into different drawer layers based on their colors."

b) Spatial-Aware Sequencing: This test suite is designed to evaluate an agent's comprehension of physical causal chains and goal instructions. Its focus lies in identifying and resolving physical obstacles to generate a coherent sequence of actions.

- **Simple**: Features two closed boxes (Red and Yellow). The Yellow box contains 5–8 random items. The agent must infer the explicit sequence of opening both lids to access and store items. *Instruction: "Transfer all items from the yellow box to the red box."*
- **General**: Features Red, Yellow, and Blue boxes in a stacked configuration where the Red box is always at the bottom. 5–8 random objects are placed on the table. The agent must analyze $OnTop$ relations to clear the stack before accessing the target box. *Instruction: "Move all objects on the table into the red box."*
- **Complex**: The scene contains 5–8 random items on table and a three-tier cabinet where the top drawer is open. The agent must recognize that the open top drawer physically obstructs access to the middle tier. *Instruction: "Transfer all objects from the box into the middle drawer."*

c) Goal-conditioned Generalization: This suite tests the agent's flexibility in translating abstract, long-horizon goals into state-aligned actions. The environmental layout is held constant with 3 colored boxes, 6–10 cubes/mugs, a milk carton, popcorn, and a book, while the instruction complexity evolves.

- **Simple**: Focuses on mapping specific unique items to distinct functional locations. *Instruction: "Place the milk, the popcorn, and the book into different drawer layers of the cabinet."*
- **General**: Focuses on attribute-based multi-object grouping and sorting. *Instruction: "Sort all cubes and mugs into the boxes of their corresponding colors."*
- **Complex**: Requires cross-asset logical mapping, aligning semantic attributes with environmental tiers. *Instruction: "Store the cubes and mugs into different drawer layers based on their colors."*